Engagement Scoring and Status Segmentation of eBook Subscribers using Exploratory Data Analysis

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Problem Statement

Digital subscription platforms run on **recurring revenue**: keeping a subscriber costs less than acquiring a new one, yet platforms treat every subscriber alike and cannot tell who is staying from who is leaving, so **generic retention campaigns** miss the readers they should reach. **Research question:** which behavioural and demographic signals distinguish high-engagement from low-engagement eBook subscribers, and what should the platform do about each group? **Phase 1** (customer file, 16,404 customers) measured subscription propensity (33.2% subscribers); **Phase 2** (engagement file, 5,455 subscribers) builds the score and the Active / Inactive / Churned status.

Literature Review

Theme	What research shows	Source	How we used it
Engagement drivers	Motivated readers show fewer, longer sessions; session depth marks engagement	Vuorinen [5]	Session minutes in the score (weight 0.25)
Churn signals	Declining usage and fast content depletion precede cancellation	Bolton [2]; Godinho [3]	Recency (0.40) and logins (0.35); status rules
Subscriber segmentation	Churn drivers differ by segment; a risk score alone mistargets offers	Rani [4]; Ascarza [8]	Active / Inactive / Churned; Low / Medium / High tiers
EDA for churn analysis	Exploration and variable selection expose the churn signal before modelling	Coussement [1]	$r = 0.88$ check; books kept out of the score
Personalisation & retention	Personalised recommendations lift retention; timed win-back beats flat discounting	Gomez-Uribe [6]; Thomas [7]	One action per status in Recommendations

Dataset

File	Rows after cleaning	Key fields
customer_data.csv	16,404 customers	demographics, income, monthly spend, subscriber flag
eBook_Engagement.csv	5,455 subscribers	logins, session minutes, books accessed, days since last activity, genre, tenure

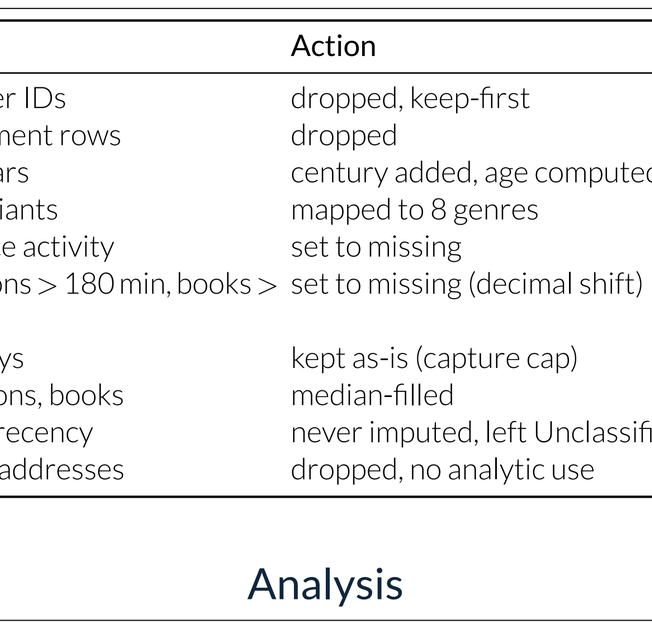
Synthetic data; the two files join on Customer ID (5,455 of 5,455 matched).

Cleaning & Preprocessing

Issue found	Action	Count
Duplicate customer IDs	dropped, keep-first	115
Duplicate engagement rows	dropped	45
Two-digit birth years	century added, age computed	16,519
Genre spelling variants	mapped to 8 genres	26
Negative days since activity	set to missing	6
Logins > 60, sessions > 180 min, books > 60	set to missing (decimal shift)	7 / 8 / 7
Recency > 365 days	kept as-is (capture cap)	11
Blank logins, sessions, books	median-filled	144
Missing or invalid recency	never imputed, left Unclassified	33
Names and street addresses	dropped, no analytic use	2 fields

Analysis

- **Score:** $100 \times (0.40R + 0.35L + 0.25S)$ on min–max scaled inputs, with R = inverted recency, L = logins, S = session minutes. **One reason per weight:** recency 0.40 is the most direct signal of leaving, logins 0.35 measure usage frequency, session minutes 0.25 measure depth of use.
- **Books Accessed excluded:** $r = 0.88$ with logins, so including it would double-count usage. Terciles at 40.80 and 51.25 give Low / Medium / High.
- **Status rules** (written and approved before computing): Active $\leq$ 30 days since last activity *and* $\geq$ 1 login/month; Inactive 31–90 days; Churned > 90 days.



Charts

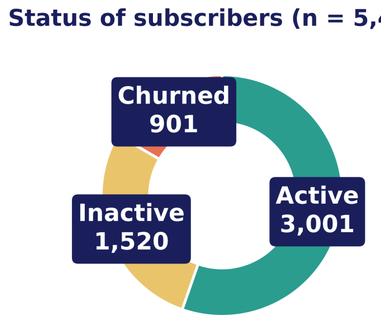


Figure 2. 55.3% Active, 28.0% Inactive, 16.6% Churned (33 unclassified).

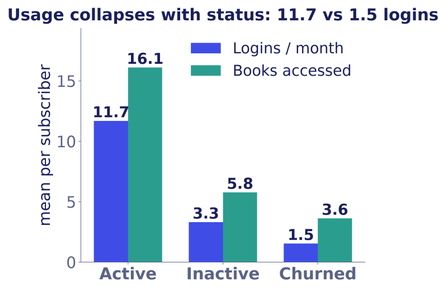


Figure 3. Churned log in 1.5 times a month vs 11.7 for Active.

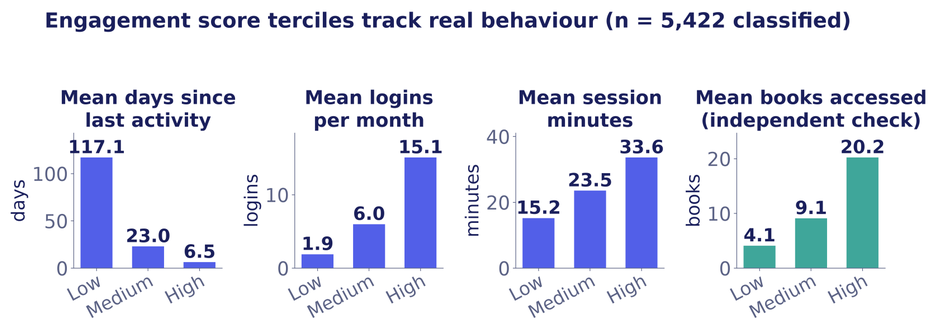


Figure 4. Books accessed – not a score input – rise 4.1 → 9.1 → 20.2 across tiers: an independent check.

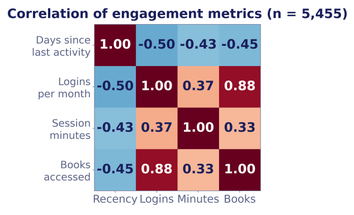


Figure 5. $r = 0.88$: books excluded.

Tables

Table 1. Status summary ($n = 5{,}455$)

Status	n	%	Median days	Mean logins	Mean books
Active	3,001	55.3	5	11.7	16.1
Inactive	1,520	28.0	58	3.3	5.8
Churned	901	16.6	152	1.5	3.6
Unclassified	33	–	–	9.9	13.1

Insights & Interpretation

- **Churned subscribers log in far less**: 11.7 vs 1.5 logins a month, and the median gap since last activity is 5 / 58 / 152 days.
- **Books accessed rise by tier** (4.1 / 9.1 / 20.2) although books are *not* a score input – the score tracks real behaviour.
- **High-value vs Churned demographics are small** (T3, merged here): spend \$94.1 vs \$83.5 ($d = 0.33$), income \$99.8k vs \$90.3k (0.23), children 2.2 vs 1.7 (0.22), age 62.2 vs 63.8 (-0.20) – all small, associations only, not causes.
- **Genre and tenure do not separate the groups**, and tier and status agree partly by construction because recency drives both.
- **Eleven Active subscribers sit in the Low tier** – an early-warning watchlist.

Recommendations

- *Active* – **retain**: personalised recommendations, and monitor the 11 Low-tier Active subscribers.
- *Inactive* – **win back before day 90**: nudge near day 30 with personalised book picks and new releases.
- *Churned* – **win-back campaign** with relevant timing and segment-specific offers, not a blanket discount.

Conclusion

- Declining logins and rising recency – not demographics – separate engagement groups; the score and status rules are reproducible.
- Limits: the study shows *who* leaves and *when*, not *why*; associations only; status is recency-based; the data is synthetic.
- Next: pilot the three strategies on real data with cancellation dates.

References

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E-mail ID: (to be added)